Number Theory & Cryptography

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Disclaimer

This document contains the lectures given by Dr. ZAIMI.

Some contents were added as remainders and extras for the students. To separate the contents of the course to actual additions or out of context information, a black band will be added by its side like the one on this comment.

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Chapter 1

Elliptic Curves

1.1. Generalities

- A plane curve is a continuous map $I \to \mathbb{R}^2, t \mapsto (x(t), y(t))$.
- A plane curve can be defined by an equation of the form $f(x, y) = 0$.
- The equation of the tangent line to the curve given by $f(x, y) = 0$ is

$$\frac{\partial f}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0)(y - y_0) = 0$$

- A plane curve is said to be algebraic if f is a polynomial.
- An algebraic curve is said to be smooth if it has a tangent line at any point.
- We can consider the field to be K with zero characteristic or finite field with characteristic not 2 or 3.

> **Definition 1.1.1 (Elliptic Curve):** *An elliptic curve E is defined as a graph of an algebraic smooth curve defined by*
>
> $$E : f(x, y) = (y^2 + a_1xy + a_3) - (x^3 + a_2x^2 + a_4x + a_6) = 0$$
>
> *with $a_1, a_2, a_3, a_4, a_6 \in K$.*

By making a change of variable, $Y = y + a_1/2x + a_3/2$ we obtain

$$Y^2 = y^2 + \left(\frac{a_1}{2}x + \frac{a_3}{2}\right)^2 + a_1xy + a_3y$$

then (E) will change to

$$Y^2 = x^3 + b_2x^3 + b_4x + b_6$$

and setting $X = x + b_2/3$

$$X^3 = x^3 + b_2x^2 + cx + d$$

replacing in (E) we obtain

$$Y^2 = X^3 + aX + b$$

which is called the reduced Weiestrass form.

> **Proposition 1.1.2:** *Let E be the curve defined by $y^2 = x^3 + ax + b$, then E is an elliptic curve $\leftrightarrow$ the cubic polynomial $x^3 + ax^2 + bx + c$ has no repeated root in K.*

Proof. Set $P(x) = x^3 + ax^2 + bx + c$, then we can rewrite the equation as $f(x, y) = y^2 - P(x)$, and hence we reformulate the statement as: the curve E is not smooth and only if $f(x, y)$ has repeated roots.

$\to$ Then $\exists (x_0, y_0) \in K^2, f(x_0, y_0) = 0$ and $\nabla f(x_0, y_0) \neq 0$, $y_0^2 = P(x_0) \to 2y_0 = 0$ and $P'(x_0) = 0$ hence P has a repeated root.

$\leftarrow$ Suppose P has a repeated root in K, then $P(x_0) = P'(x_0) = 0$, set $y_0 = 0$, $(x_0, y_0) \in E$ because $P(x_0) = 0 = y_0^2$, we obtain that the partial derivatives of f are 0 and hence E is not a smooth curve. ■

> **Proposition 1.1.3:** *Let $P(x) = x^3 + ax^2 + bx + c \in K[X]$, then P has a repeated root in $K \leftrightarrow P$ has a repeated root in $\overline{K}$.*

Proof. $\to$ trivial.

$\leftarrow$ Suppose that P has a repeated root in $\overline{K}$, say $x_0 \in \overline{K}$, $P(x_0) = P'(x_0) = 0$, then x_0 is a root of $P'(x) = 3x^2 + 2ax + b$, if $\deg_K(x_0) = 1$, done. Otherwise, if $\deg_K(x_0) = 2$, then $P'(x) = 3\,\mathrm{Irr}(x_0, K, x)$, write $P(x) = (x - x_0)^2(x - x_1)$ in $\overline{K}[X]$ then $P(x) = \mathrm{Irr}(x_0, K, x)(x - x_2)$ with $x_1, x_2 \in \overline{K}$ then $x - x_0 \in K[X] \to x_2 \in K$, hence we obtain

$$\begin{aligned} P(x) &= (x - x_0)(x - x_0)(x - x_1) \\ &= (x - x_0)(x - \tilde{x}_0)(x - x_0) \end{aligned}$$

■

Definition 1.1.4 (Polynomial Descriminant): *Let $P(X) \in R[X]$ monic, $P(X) = \prod_{i=1}^{n}(X - \alpha_i)$, the discriminant of P is defined as*

$$\operatorname{disc}(P) = \prod_{1 \leq i < j \leq n} (\alpha_i - \alpha_j)^2$$

For $P(x) = x^3 + ax + b$, we have $\operatorname{disc}(P) = -4a^3 - 27b^2$. It follows that P has no repeated root in $\overline{K} \leftrightarrow \operatorname{disc}(P) \neq 0$.

Corollary 1.1.5: *An algebraic curve $E : y^2 = x^3 + ax + b$ is an elliptic curve if and only if $4a^3 + 27b^2 \neq 0$.*

Definition 1.1.6 (Elliptic Curve Discriminant): *Let $E : y^2 = x^3 + ax + b$ be an elliptic curve, then the discriminant of E is defined as $\Delta(E) = 16 \operatorname{disc}(x^3 + ax + b)$.*

Definition 1.1.7 (J-Invariant): *Let $E : y^2 = x^3 + ax + b$ be an elliptic curve, $a, b \in K$, then the J-invariant of E is given by*

$$J(E) = 1728 \frac{4a^3}{4a^3 + 27b^2}$$

Proposition 1.1.8: *Consider the two following elliptic curves over K*

$$E_1 : y^2 = x^3 + a_1 x + b_1 \qquad E_2 : y^2 = x^3 + a_2 x + b_2$$

$J(E_1) = J(E_2)$ if and only if $\exists \mu \in \overline{K}^*, a_2 = \mu^4 a_1$ and $b_2 = \mu^6 b_1$.